

Project Title

Assessing the Effectiveness of Cookie Consent Systems: An Automated Analysis of GDPR Compliance and Privacy Risks on Popular Websites

2. Team Members

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3. Project Overview

Background

Websites use cookies and trackers to gather user data for personalization, advertising, and analytics. Under the EU's GDPR (in force since 2018), sites must get explicit, informed consent before placing non-essential cookies—meaning users should have clear accept/reject choices, no manipulative “dark patterns,” and no tracking activated before consent. In practice, enforcement is uneven and many sites may bypass these requirements, increasing privacy risks like unauthorized data collection and inferring sensitive information. This project builds an automated system to test real websites' cookie consent banners and measure both GDPR compliance and how well they actually protect users' privacy.

Project Type

This is an **implementation-based project**, emphasizing the development of technical tools for data collection, analysis, and reporting. It avoids purely theoretical or legal discussions, instead building a functional pipeline to measure real-world privacy compliance.

4. Objectives

Primary Question

Do cookie consent systems on popular websites effectively protect user privacy under GDPR requirements?

Specific Goals

- Evaluate compliance with GDPR consent rules (e.g., presence of reject options, no pre-consent tracking).
- Quantify privacy risks, such as the deployment of tracking cookies before user interaction.
- Identify manipulative UI elements (dark patterns) that may undermine informed consent.
- Generate actionable insights, including metrics, visualizations, and a compliance report.

5. Methodology

System Architecture and Pipeline

We will design and implement a modular system using browser automation and data analysis tools. The architecture follows a pipeline model:

1. Website Selection and Data Collection:

- Compile a list of 100-200 websites from sources like Alexa Top Sites or SimilarWeb rankings, categorized by type (e.g., e-commerce, news, social media) and region (focusing on EU-targeted sites).
- Use browser automation to simulate user visits from an EU IP (via VPN or proxy) to trigger GDPR-compliant banners.
- Tools: Python with Selenium or Playwright for headless browsing; libraries like Selenium Wire or mitmproxy for capturing network traffic and cookies.

2. Cookie and Tracker Capture:

- Visit each site without interacting with the consent banner.
- Record all cookies set, HTTP requests, and trackers (e.g., via browser storage inspection).
- After capture, simulate consent (accept/reject) and compare post-consent state.
- Integrate detection for dark patterns (e.g., analyze banner HTML for button visibility, size, or click requirements) and compliance (e.g., check for "Reject All" button).

3. Analysis and Metrics Computation:

- Classify cookies/trackers using databases like EasyList or WhoTracksMe (via API or local parsing).
- Metrics:
 - Percentage of sites with pre-consent trackers (% violations).
 - Average number of trackers per site (before/after consent).
 - Tracker types and vendors (e.g., Google Analytics, Meta Pixel).
 - Dark pattern prevalence (e.g., % with asymmetric buttons).
 - Compliance score (0-100) based on GDPR criteria (e.g., explicit consent, easy rejection).
- Handle edge cases: JavaScript obfuscation, anti-bot measures (via delays or user-agent rotation).

4. Reporting and Visualization: (this is for the report we can omit it too)

- Generate graphs (e.g., bar charts for violation rates by category, pie charts for tracker vendors) using Matplotlib or Seaborn.
- Produce a GDPR compliance report in PDF/HTML format, ranking sites and highlighting examples.
- Tools: Pandas for data processing, Jupyter Notebooks for prototyping.

Privacy-Enhancing Technologies (PETs) Integration

- We will apply basic PETs in our tool, such as anonymizing collected data and ensuring ethical scraping (respecting robots.txt, rate-limiting requests).
- Evaluate defenses: Test if rejecting consent effectively blocks trackers, relating to broader PETs like differential privacy in aggregated reporting.

6. References

- GDPR Regulation (EU) 2016/679.
- CNIL Guidelines on Cookies and Trackers.
- Prior studies (e.g., Nouwens et al., "Dark Patterns after the GDPR").